

The Cytoskeleton in Morphogenesis

In animals, movement of parts of a cell can bring about changes in cell shape or enable a cell to migrate from one place to another within the embryo. One set of cellular components essential to these events is the collection of microtubules and microfilaments that make up the cytoskeleton (see Table 6.1).

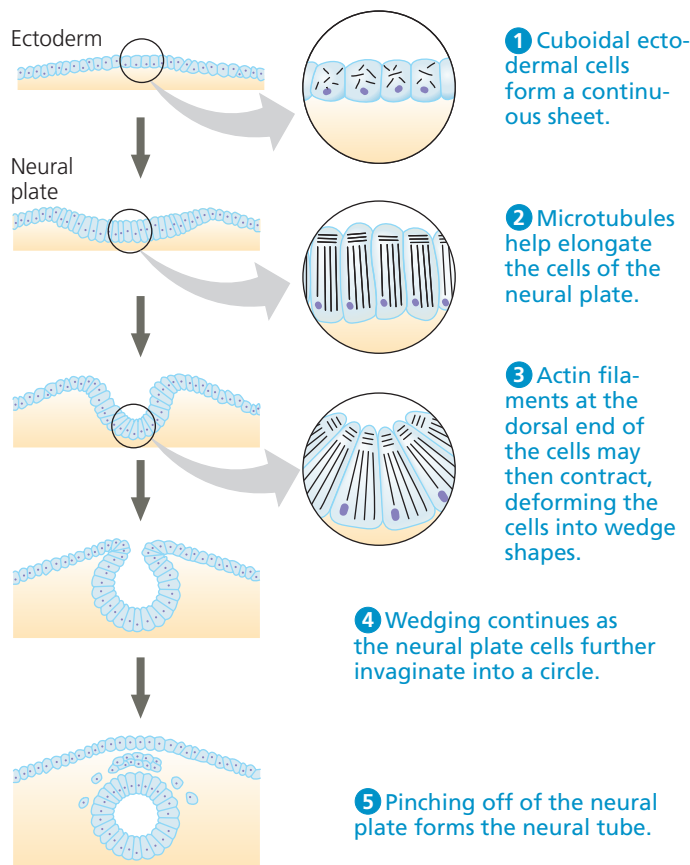
Cell Shape Changes in Morphogenesis

Reorganization of the cytoskeleton is a major force in changing cell shape during development. As an example, let's return to the topic of neurulation. At the onset of neural tube formation, microtubules oriented from dorsal to ventral in a sheet of ectodermal cells help lengthen the cells along that axis (**Figure 47.16**). At the apical end of each cell is a bundle of actin filaments (microfilaments) oriented crosswise. These actin filaments contract, giving the cells a wedge shape that bends the ectoderm layer inward.

The generation of wedge-shaped cells by apical constriction of actin filaments is a common mechanism in development for invaginating a cell layer. For instance, during gastrulation in the fruit fly *Drosophila melanogaster*, the formation of wedge-shaped cells along the ventral surface drives generation of the tube of cells that forms the mesoderm.

▼ Figure 47.16 Change in cell shape during morphogenesis.

Reorganization of the cytoskeleton is associated with morphogenetic changes in embryonic tissues, as shown here.



The cytoskeleton also directs a morphogenetic movement called **convergent extension**, a rearrangement that causes a sheet of cells to become narrower (converge) while it becomes longer (extends). This type of lengthening and narrowing of cells occurs often in gastrulation, including formation of the primitive streak in the fertilized chick egg (see Figure 47.11) and elongation of the archenteron in the sea urchin embryo (see Figure 47.8). Convergent extension is also important during involution in the frog gastrula. There, convergent extension changes the gastrulating embryo from a spherical shape to the rounded rectangular shape seen in Figure 47.14c.

The cell movements during convergent extension are quite simple: The cells elongate, with their ends pointing in the direction they will move, and then wedge between each other, forming fewer columns of cells (**Figure 47.17**). It's like a crowd of people about to enter a theater moving forward while also merging into a single-file line.

Cell Migration in Morphogenesis

The cytoskeleton is responsible not only for cell shape changes but also for cell migration. During organogenesis in vertebrates, cells from the neural crest and from somites migrate to locations throughout the embryo. Cells “crawl” within the embryo by using cytoskeletal fibers to extend and retract cellular protrusions. This type of motility is akin to amoeboid movement (see Figure 6.26b). Transmembrane glycoproteins called *cell adhesion molecules* play a key role in cell migration by promoting interaction between pairs of cells. Cell migration also involves the *extracellular matrix (ECM)*, the meshwork of secreted glycoproteins and other macromolecules lying outside the plasma membranes of cells (see Figure 6.28).

The ECM helps to guide cells in many types of movements, such as migration of individual cells and shape changes of cell sheets. Cells that line migration pathways regulate the movement of migrating cells by secreting specific molecules into the ECM. For these reasons, researchers are attempting to generate an artificial ECM that can serve as a scaffold for the repair or replacement of damaged tissues or organs. One promising approach involves the use of nanofiber fabrication to produce materials that mimic the essential properties of the natural ECM.

Programmed Cell Death

Just as certain cells of the embryo are programmed to change shape or location, others are programmed to die. At various times in development, individual cells, sets of cells, or whole tissues cease to develop, die, and are engulfed by neighboring cells. Thus, *programmed cell death*, also called **apoptosis**, is a common feature of animal development.

One circumstance for programmed cell death is when a structure functions only in a larval or other immature form of the organism. One familiar example is the tail of a tadpole, the free-swimming larval stage of a frog or toad. The tail forms during early development, enables locomotion during